

# Mixture of Flows

Notes on Multi-Teacher Diffusion Distillation  
via Learnable (Teacher×Source×Time) Routing

Flow Factory

## Abstract

We study the problem of distilling  $K$  specialized diffusion teachers  $\{v^{(k)}\}_{k=1}^K$  (each fine-tuned for a different reward objective) into a single student that achieves *teacher-level in-domain performance on every reward* while picking up cross-objective gains on out-of-distribution (OOD) rewards. We make three observations:

(i) Standard On-Policy Distillation (OPD) for diffusion decomposes into a path-wise MSE term and a policy-gradient (PG) term; the PG term has zero contribution in expectation under the teacher-as-target setting, so *plain velocity MSE is sufficient*.

(ii) Naive multi-teacher OPD has two natural failure modes: *route-by-source* caps the student at single-teacher performance with no OOD gain, and *averaging* loses in-domain by Jensen’s inequality with only a small OOD bonus.

(iii) We can construct a *potentially better target* by taking a learnable convex combination  $v_\lambda = \sum_k \lambda_{k,t,s,c} v^{(k)}$  with weights conditioned on (teacher, time, source, prompt). Because the convex combination of valid flow-matching velocities is itself a valid flow-matching velocity, this  $v_\lambda$  is a legitimate diffusion model and can be tuned *by RL* (DiffusionNFT or Flow-GRPO) on the small mixing parameter space. The student is then distilled from  $v_{\lambda^*}$  via plain pathwise MSE.

The framework provides a *constructive* Level-1 in-domain guarantee: the student’s expected reward on every in-domain dataset is at least that of the corresponding single teacher. We also establish that strict OOD improvement is *not* guaranteed in general; a precise negative theorem with three formal obstructions is given, alongside three sufficient conditions (temporal complementarity, non-conflicting reward gradients, non-empty Pareto-improving region) under which OOD gains become provable. We close with the experimental program (three teachers trained to GenEval=0.958, OCR=0.953, PickScore≈24.13) and a list of failure modes observed during preliminary runs.

## Contents

|          |                                                                                        |          |
|----------|----------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                                                    | <b>2</b> |
| 1.1      | Motivation . . . . .                                                                   | 2        |
| 1.2      | Failure modes of prior approaches . . . . .                                            | 2        |
| 1.3      | Our approach . . . . .                                                                 | 2        |
| 1.4      | Contributions and roadmap . . . . .                                                    | 2        |
| <b>2</b> | <b>Background</b>                                                                      | <b>3</b> |
| 2.1      | Flow matching and notation . . . . .                                                   | 3        |
| 2.2      | OPD: pathwise vs. policy-gradient . . . . .                                            | 3        |
| 2.3      | Convex combinations of flow matching velocities are flow-matching velocities . . . . . | 4        |
| 2.4      | Failure modes of multi-teacher OPD . . . . .                                           | 4        |
| <b>3</b> | <b>Method: Mixture of Flows</b>                                                        | <b>4</b> |
| 3.1      | The MoF construction . . . . .                                                         | 4        |
| 3.2      | Two-stage pipeline . . . . .                                                           | 5        |

|          |                                                            |           |
|----------|------------------------------------------------------------|-----------|
| 3.3      | Algorithmic choices . . . . .                              | 5         |
| 3.3.1    | NFT vs. Flow-GRPO . . . . .                                | 5         |
| 3.3.2    | LUT vs. Router . . . . .                                   | 6         |
| <b>4</b> | <b>Theoretical Analysis</b>                                | <b>6</b>  |
| 4.1      | Level-1 in-domain guarantee (constructive) . . . . .       | 6         |
| 4.2      | Level-2 conditional in-domain strict-improvement . . . . . | 6         |
| 4.3      | No general OOD strict-improvement guarantee . . . . .      | 7         |
| 4.4      | NFT $\beta$ -cancellation . . . . .                        | 7         |
| 4.5      | Distillation gap . . . . .                                 | 8         |
| <b>5</b> | <b>Experimental Program</b>                                | <b>8</b>  |
| 5.1      | Teacher training (preparation, completed) . . . . .        | 8         |
| 5.2      | MoF configurations . . . . .                               | 8         |
| 5.3      | Evaluation . . . . .                                       | 9         |
| <b>6</b> | <b>Empirical findings and ablation plan</b>                | <b>9</b>  |
| 6.1      | Pathwise vs. PG (verification of Prop. 1) . . . . .        | 9         |
| 6.2      | Naive multi-teacher OPD baselines (Step 2) . . . . .       | 9         |
| 6.3      | MoF main: NFT vs. Flow-GRPO (Step 3) . . . . .             | 10        |
| 6.4      | Temporal analysis of $\lambda^*$ . . . . .                 | 10        |
| 6.5      | PickScore-dominance failure mode in NN routers . . . . .   | 10        |
| 6.6      | $\gamma$ ablation . . . . .                                | 10        |
| 6.7      | Initialization ablation . . . . .                          | 11        |
| <b>7</b> | <b>Discussion</b>                                          | <b>11</b> |
| 7.1      | The asymmetry of in-domain vs. OOD . . . . .               | 11        |
| 7.2      | When to use which architecture . . . . .                   | 11        |
| 7.3      | Limitations . . . . .                                      | 11        |
| <b>8</b> | <b>Conclusion</b>                                          | <b>12</b> |
| 8.1      | Open problems . . . . .                                    | 12        |
| <b>A</b> | <b>Pointers to supporting documents</b>                    | <b>12</b> |

# 1 Introduction

## 1.1 Motivation

Modern text-to-image diffusion fine-tuning has become a multi-objective problem: human preference (*e.g.* PickScore), compositional accuracy (*e.g.* GenEval), and text-rendering fidelity (OCR) are all desirable, each with its own reward model and training set. RL fine-tuning (Flow-GRPO, DiffusionDPO, DiffusionNFT) typically optimizes a single reward, so the practical pipeline is to train  $K$  *specialist* LoRA teachers  $\{v^{(k)}\}_{k=1}^K$  separately—producing strong in-domain performance per reward, but no single deployable artifact that excels at all of them.

The natural follow-up question is: *can we distill these  $K$  teachers into a single student*, with the student matching each teacher’s in-domain strength while picking up cross-objective gains? We refer to in-domain performance as the score of reward  $R_s$  on prompt set  $\mathcal{P}_s$  (the set the teacher  $v^{(s)}$  was trained on), and OOD performance as  $R_j$  on  $\mathcal{P}_s$  for  $j \neq s$ .

## 1.2 Failure modes of prior approaches

Two natural baselines, both descended from On-Policy Distillation (OPD), fail in symmetric ways:

- **Route-by-source.** Each prompt is routed to its source-matched teacher and the student fits only that teacher. The student’s performance is upper-bounded by the corresponding single teacher; *no* OOD gain because cross-teacher information never enters the gradient.
- **Averaging.** The student fits the arithmetic mean  $\bar{v} = \frac{1}{K} \sum_k v^{(k)}$ . By Jensen’s inequality (and the  $\alpha$ -relation between velocity-MSE and KL distillation), the student’s in-domain performance *strictly degrades* below the single-teacher baseline; only marginal OOD bonus is observed.

Both modes leak from the same fundamental constraint: the student’s velocity field is rigidly fixed at every  $(x, t)$  once the target is specified. There is no slack to spend on *cross-objective synergy*, where one teacher might be best at  $R_s$  in early timesteps (layout) and another teacher best at  $R_j$  in late timesteps (texture).

## 1.3 Our approach

We build a *better teacher* before distilling. Specifically, let

$$v_\lambda(x, t, c) = \sum_{k=1}^K \lambda_k(t, s(c), c) v^{(k)}(x, t, c), \quad \sum_k \lambda_k = 1, \quad \lambda_k \geq 0, \quad (1)$$

with weights  $\lambda$  depending on the teacher index  $k$ , the timestep  $t$ , the source label  $s(c)$ , and possibly the prompt embedding  $c$ . We optimize the  $\lambda$ ’s (and only those) by RL on a composite reward *in the velocity-output space*, then distill the final  $v_{\lambda^*}$  into a single student LoRA via plain pathwise MSE.

## 1.4 Contributions and roadmap

1. **Methodological** (§3): A three-stage pipeline—OPD pathwise sufficiency  $\rightarrow$  MoF target construction by RL  $\rightarrow$  on-policy distillation—decoupling “find a good convex weighting” from “encode it in a single LoRA”. The mixing search space is just  $K \times T \times S \sim 10^2$  parameters (LUT) or  $\sim 10^6$  (router), tiny compared to the  $10^7$ – $10^9$  student parameters.

2. **Theoretical** (§4): *Constructive* Level-1 in-domain guarantee (§4.1); *conditional* Level-2 in-domain strict-improvement under temporal complementarity (§4.2); a *negative* theorem on OOD strict improvement (§4.3) showing the asymmetry between in-domain and OOD, accompanied by three sufficient conditions; an analysis of NFT’s  $\beta$ -cancellation property (§4.4); and a discussion of distillation-gap erosion (§4.5).
3. **Experimental** (§5): A protocol with three Flow-DPPO teachers (GenEval/OCR/PickScore at 0.958/0.953/24.128 respectively), four mixing-module variants (LUT, LUT-SA, Time Router, adaLN/MLP Router), two RL options (DiffusionNFT, Flow-GRPO), and ablations on the OOD-bonus  $\gamma$  and initialization. We report a preliminary failure mode—the *PickScore-dominance* effect—in which neural routers asymmetrically inflate PickScore signal because PickScore’s `applicable_sources` covers all sources while the others are domain-specific.

## 2 Background

### 2.1 Flow matching and notation

Let  $x_0 \sim p_{\text{data}}$ ,  $x_1 \sim \mathcal{N}(0, I)$ . Flow matching parameterizes a velocity field  $v_\theta(x, t, c)$  such that the ODE  $\dot{x}_t = v_\theta(x_t, t, c)$  transports the noise distribution to the data distribution. The standard training loss is

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{x_0, x_1, t} \left[ \|v_\theta(x_t, t, c) - (x_1 - x_0)\|_2^2 \right], \quad x_t = (1 - t)x_0 + tx_1. \quad (2)$$

A teacher is a frozen  $v^{(k)}$  (typically a LoRA snapshot of a base flow model). We index teachers by  $k \in \{1, \dots, K\}$ , sources by  $s \in \{1, \dots, S\}$  (where each source has an in-domain reward  $R_s$ ), timesteps by  $i \in \{0, \dots, T - 1\}$ , and per-source prompt sets by  $\mathcal{P}_s$ . We use  $f_j^{(s)}(\lambda) \triangleq \mathbb{E}_{p \sim \mathcal{P}_s} [R_j(\text{ODE}(v_\lambda; x_T, p))]$  to denote the expected reward  $R_j$  achieved by mixing  $\lambda$  on prompt set  $\mathcal{P}_s$ .

### 2.2 OPD: pathwise vs. policy-gradient

Suppose the goal is to distill a teacher  $v^{\text{tgt}}$  into a student  $v_\theta$ . Let  $q_\theta$  be the student’s induced trajectory distribution and  $q_{\text{tgt}}$  the teacher’s. Minimizing the path-wise KL  $\text{KL}(q_\theta \| q_{\text{tgt}})$  has the well-known decomposition (cf. standard OPD literature):

$$\nabla_\theta \text{KL}(q_\theta \| q_{\text{tgt}}) = \underbrace{\mathbb{E}_{q_\theta} \left[ \sum_i \frac{1}{2\sigma_i^2} \nabla_\theta \|\mu_\theta^i - \mu_{\text{tgt}}^i\|^2 \right]}_{\text{pathwise}} + \underbrace{\mathbb{E}_{q_\theta} \left[ \sum_i (\log q_\theta - \log q_{\text{tgt}}) \nabla_\theta \log q_\theta \right]}_{\text{policy-gradient (REINFORCE)}}. \quad (3)$$

**Proposition 1** (PG term has zero contribution in expectation). Under standard assumptions (smooth densities, finite variance) and when  $v_\theta$  is parameterized so that  $\mathbb{E}_{q_\theta} [\nabla_\theta \log q_\theta] = 0$  at every intermediate step (score-function identity), the policy-gradient term in (3) is mean-zero *conditionally on the pathwise gradient already being applied*, contributing only variance.

*Proof sketch.* Define  $r_i := \log q_\theta - \log q_{\text{tgt}}$  and  $g_i := \nabla_\theta \log q_\theta$ . By the score-function identity  $\mathbb{E}[g_i] = 0$ . If  $r_i$  and  $g_i$  are independent (or weakly correlated) under  $q_\theta$ , then  $\mathbb{E}[r_i g_i] \approx \mathbb{E}[r_i] \mathbb{E}[g_i] = 0$ . In the diffusion setting where the sampling trajectory is generated by  $v_\theta$ , the pathwise term already corrects  $\mu_\theta^i$  toward  $\mu_{\text{tgt}}^i$ , so any *remaining* REINFORCE signal is high-variance noise rather than a directional bias. Empirically (and as established in the OPD literature), training with the pathwise term alone reaches the same fixed point as training with the full KL gradient.  $\square$

**Practical consequence:** distillation in our pipeline reduces to plain velocity MSE (no log-prob, no importance ratio, no clipping). This is what `trainers/mof/distill.py` implements (see `mof_two_stage_pipeline.tex`).

## 2.3 Convex combinations of flow matching velocities are flow-matching velocities

**Proposition 2** (Convex closure of FM). If  $v^{(1)}, \dots, v^{(K)}$  are valid flow-matching velocity fields (each satisfying the continuity equation  $\partial_t p_t + \nabla \cdot (p_t v^{(k)}) = 0$ ), then for any  $\lambda \in \Delta^{K-1}$ ,

$$v_\lambda = \sum_k \lambda_k v^{(k)} \quad (4)$$

is also a valid flow-matching velocity field.

*Proof.* The continuity equation is linear in  $v$ ; convex combinations of solutions remain solutions. (Formally, the corresponding marginal density is the convex combination of teacher marginals.)  $\square$

This is the key structural fact that enables MoF: *any* per-timestep mixing  $v_\lambda$  defines a legitimate diffusion model whose ODE is well-posed and whose trajectories can be sampled, scored, and distilled.

## 2.4 Failure modes of multi-teacher OPD

Let  $f_j^{(s)}(\theta)$  denote the student’s expected reward  $R_j$  on prompt set  $\mathcal{P}_s$  after distillation.

**Route-by-source.** Train  $K$  separate students or a single student that switches teacher by  $s(p)$ . The optimal student velocity is  $v_\theta^*(x, t | p \in \mathcal{P}_s) = v^{(s)}(x, t)$ , hence  $f_s^{(s)}(\theta^*) \approx f_s^{(s)}(v^{(s)})$  (in-domain matches teacher exactly, modulo distillation gap), and  $f_j^{(s)}(\theta^*) \approx f_j^{(s)}(v^{(s)})$  (OOD just inherits teacher  $s$ ’s OOD score—no cross-teacher transfer). *No OOD gain.*

**Averaging.** Train a student to fit  $\bar{v} = \frac{1}{K} \sum_k v^{(k)}$ . The optimal student is  $v_\theta^* = \bar{v}$ . By Jensen’s inequality applied to sufficiently sharply-peaked rewards,  $f_s^{(s)}(\bar{v}) < f_s^{(s)}(v^{(s)})$  for every in-domain pair  $(s, s)$  (*strict in-domain degradation*). OOD often improves slightly because  $\bar{v}$  blends teacher behaviors, but the gain is small and not in general guaranteed.

Both failure modes share a structural cause

The student’s optimal velocity is rigidly determined by the target construction: route picks a vertex of the teacher convex hull ( $v^{(s)}$ ), averaging picks the centroid. Neither uses the per-timestep slack that Proposition 2 grants. To exploit it, the weights must be *learned* based on what actually helps the reward landscape—this is the MoF idea.

# 3 Method: Mixture of Flows

## 3.1 The MoF construction

Let

$$v_\lambda(x, t, c) = \sum_{k=1}^K \lambda_k(t, s(c), c; \psi) v^{(k)}(x, t, c), \quad \lambda_k = \frac{\exp(\ell_k/\tau)}{\sum_{k'} \exp(\ell_{k'}/\tau)}. \quad (5)$$

Here  $\psi$  is a small set of *learnable mixing parameters* (logits  $\ell$  shaped through softmax). The teachers are frozen and detached.

We instantiate  $\lambda$  in five flavors of increasing expressiveness, each realized in `trainers/mof/common.py` by a distinct `MoFMixingModule` subclass:

| Module       | Form of $\ell$                                         | Notes                                                               |
|--------------|--------------------------------------------------------|---------------------------------------------------------------------|
| LUT          | $\ell \in \mathbb{R}^{K \times T \times S}$            | Per-source & per-step lookup; $K \cdot T \cdot S \sim 10^2$ params. |
| LUT-SA       | $\ell \in \mathbb{R}^{K \times T}$                     | Source-agnostic; broadcast over $S$ . Diagnostic baseline.          |
| Time Router  | $\ell = \text{MLP}(t)$                                 | Continuous in $t$ ; no prompt conditioning.                         |
| adaLN Router | $\ell = \text{MLP}(\text{adaLN}(t, c))$                | adaLN-fused (DiT-style).                                            |
| MLP Router   | $\ell = \text{MLP}([\text{TE}(t) \  c_{\text{pool}}])$ | Concat-then-MLP.                                                    |

We collectively denote the parameter set  $\psi$  (whether  $K \times T \times S$  logits or router weights). The number of parameters in  $\psi$  is at least four orders of magnitude smaller than the number of student parameters.

### 3.2 Two-stage pipeline

**Stage 1 (RL on  $\psi$ ).** We optimize  $\psi$  by RL on a composite reward, treating  $v_\lambda$  as the policy. Because  $v_\lambda$  is a valid FM (Prop. 2), any RL algorithm that supports a flow-matching policy applies. We use **DiffusionNFT** as the primary algorithm (path-wise reconstruction loss reweighted by advantage; details in §3.3.1 and `mof_two_stage_pipeline.tex`). Alternative: **Flow-GRPO** (PPO-style ratio with a stochastic CPS scheduler) is supported as an ablation.

**Stage 2 (distill  $v_{\lambda^*}$  into a student LoRA).** After Stage 1,  $\psi^*$  is frozen and defines the *router-induced teacher*  $v^{\text{MoF}^*}(x, t, c) = \sum_k \lambda_k(t, s(c), c; \psi^*) v^{(k)}(x, t, c)$ . We train a single student LoRA  $\theta$  by on-policy MSE:

$$\mathcal{L}_{\text{dist}}(\theta) = \mathbb{E}_{(x_t, t, c) \sim q_\theta} \left[ \|v^\theta(x_t, t, c) - v^{\text{MoF}^*}(x_t, t, c)\|_2^2 \right], \quad (6)$$

where  $q_\theta$  is the student’s own trajectory distribution (Stage 2 is on-policy: the student rolls out, then at each visited  $(x_t, t)$  the teacher mixture is computed and matched). Because Prop. 1 eliminates the PG term, this MSE is the entire distillation loss.

**Why two stages?** Stage 1 finds  $\psi^*$  in a tiny parameter space using small batches of RL signal; Stage 2 amortizes the  $K$ -teacher inference cost into a single LoRA forward. At deployment, the student has the same inference cost as any single teacher, while behaving as the optimized convex combination.

### 3.3 Algorithmic choices

#### 3.3.1 NFT vs. Flow-GRPO

DiffusionNFT optimizes  $\psi$  via reward-weighted self-normalized MSE on  $x_0$  reconstruction:

$$\mathcal{L}_{\text{NFT}} = \frac{r \cdot \mathcal{M}(v^+) + (1 - r) \cdot \mathcal{M}(v^-)}{\beta} \cdot A_{\text{max}}, \quad \mathcal{M}(v) = \frac{\|x_t - \sigma v - x_0\|^2}{\text{sg}(\|x_t - \sigma v - x_0\|_1/d)}, \quad (7)$$

where  $r = \text{clip}(\frac{A}{2A_{\text{max}}} + \frac{1}{2}, 0, 1)$  is the normalized advantage,  $v^+ = \beta v_{\text{new}} + (1 - \beta)v_{\text{old}}$  and  $v^- = (1 + \beta)v_{\text{old}} - \beta v_{\text{new}}$  are positive/negative interpolated velocities, and  $v_{\text{old}}$  is the EMA-shadow mixing (off-policy). Detailed derivation: `mof_loss_gradient_analysis.tex`.

Flow-GRPO instead uses the PPO-clipped log-prob ratio  $\rho = \exp(\log p_{\text{new}} - \log p_{\text{old}})$ , requires a stochastic dynamics (CPS / Flow-SDE), and is on-policy in the strict sense.

**Choice:** We use NFT as the default. NFT *converges faster* empirically on the small-parameter MoF search (consistent with its self-normalized weighting; the gradient signal is large at the start and self-attenuates as  $x_0^{\text{pred}} \rightarrow x_0$ ). NFT also doesn’t require log-prob computations, halving the memory footprint of Stage 1.

### 3.3.2 LUT vs. Router

LUT-style modules give *decoupled* per-source per-timestep weights ( $\Lambda_{k,t,s}$ ), making the optimization landscape  $T \cdot S$  independent simplex problems. Routers give continuous  $t$ -flexibility (decouple training and inference  $T$ ) and prompt conditioning, but introduce a parameter-sharing bias whose pitfalls we discuss in §6.

## 4 Theoretical Analysis

We work in the per-set MoF setting: each  $\mathcal{P}_s$  has its own  $\psi_s$  (or its slice  $\Lambda_{:, :, s}$ ), so the optimization decouples across sources.

### 4.1 Level-1 in-domain guarantee (constructive)

**Theorem 1** (In-domain  $\geq$  single teacher). Let  $\psi_s^*$  be a globally-optimal Stage-1 mixing for prompt set  $\mathcal{P}_s$  under any composite reward of the form  $F^{(s)}(\lambda) = f_s^{(s)}(\lambda) + \gamma \sum_{j \neq s} f_j^{(s)}(\lambda)$  with  $\gamma \geq 0$ . Then  $\delta_s^{\text{const}}$  (assigning weight 1 to teacher  $s$  at every timestep) is a feasible point and:

$$F^{(s)}(\psi_s^*) \geq F^{(s)}(\delta_s^{\text{const}}) \geq f_s^{(s)}(v^{(s)}) - \gamma \cdot |\Delta_{\text{OOD}}|, \quad (8)$$

*i.e.* the in-domain reward of  $v_{\lambda_s^*}$  is at least the in-domain reward of teacher  $s$ , up to a small slack  $\gamma|\Delta_{\text{OOD}}|$  that vanishes as  $\gamma \rightarrow 0$ .

*Proof.*  $\delta_s^{\text{const}} \in (\Delta^{K-1})^T$  is in the search space. Its induced velocity is exactly  $v^{(s)}$ , so  $f_s^{(s)}(\delta_s^{\text{const}}) = f_s^{(s)}(v^{(s)})$ . The inequality follows from optimality of  $\psi_s^*$  on  $F^{(s)}$ .  $\square$

This is the *key advantage* of MoF over OPD-Average: the optimal policy never has to drop below single-teacher in-domain quality.

### 4.2 Level-2 conditional in-domain strict-improvement

**Theorem 2** (Strict in-domain improvement under temporal complementarity). Suppose for some timestep  $i^*$  and teacher  $k^* \neq s$ ,  $\partial f_s^{(s)} / \partial \lambda_{k^*, i^*} |_{\lambda = \delta_s^{\text{const}}} > 0$  (teacher  $k^*$ ’s velocity at step  $i^*$  contributes positively to  $R_s$ ). Then there exists a neighborhood of  $\delta_s^{\text{const}}$  inside  $(\Delta^{K-1})^T$  such that  $f_s^{(s)}(\lambda) > f_s^{(s)}(\delta_s^{\text{const}})$  on a positive-measure subset, and consequently  $\psi_s^*$  achieves *strict* in-domain improvement.

**Remark 1.** This condition—“in some timestep, mixing in another teacher actually helps the in-domain reward”—is what we call **temporal complementarity**. It is an *empirical* property of the teacher ensemble and reward landscape, not a structural property of MoF. Whether it holds depends on whether teachers truly specialize at different denoising stages.

### 4.3 No general OOD strict-improvement guarantee

The natural follow-up question is: does MoF guarantee that the student *strictly exceeds* every single teacher on every OOD reward?

**Theorem 3** (No general OOD guarantee). There exist legitimate MoF instances  $(K, T, \{v^{(k)}\}, \{R_j\}, \mathcal{P}_s, \gamma)$  for which, at any optimum  $\psi_s^*$ , some OOD reward  $R_j$  ( $j \neq s$ ) satisfies

$$f_j^{(s)}(\psi_s^*) < \rho_j^{(s), \text{const}} \triangleq \max_k f_j^{(s)}(\delta_k^{\text{const}}). \quad (9)$$

Thus no framework-only argument (without additional structural assumptions) can prove strict OOD improvement.

#### Three formal obstructions.

1. **Vertex-maximum block.** If  $f_j^{(s)}$  is quasi-concave on  $(\Delta^{K-1})^T$  with maximum at a single *constant* vertex  $\delta_{k^*}^{\text{const}}$ , then no convex combination can exceed  $f_j^{(s)}(\delta_{k^*}^{\text{const}}) = \rho_j^{(s), \text{const}}$ .
2. **Composite misalignment.**  $\psi_s^* = \arg \max F^{(s)}$  generally satisfies  $\nabla f_j^{(s)}(\psi_s^*) \neq 0$ , so  $\psi_s^*$  is *not* a critical point of  $f_j^{(s)}$ . The composite optimum gives up some  $f_j$  to gain  $f_s + \sum_{j' \neq j} f_{j'}$ .
3. **Reward conflict.** If  $\nabla f_s^{(s)} \cdot \nabla f_j^{(s)} < 0$  near  $\delta_s^{\text{const}}$ , the composite optimum lies on a Pareto front and cannot strictly beat both extremes.

**Sufficient conditions for OOD strict improvement.** Under any one of the following, the negative theorem can be strengthened to a positive guarantee:

- (A) *Temporal complementarity for OOD:* different teachers maximize  $\partial f_j^{(s)} / \partial \lambda_{k,i}$  at different timesteps  $i$ , AND a Pareto-improving region  $\mathcal{Q}_j = \{\lambda : f_s^{(s)}(\lambda) \geq f_s^{(s)}(\delta_s^{\text{const}}) \text{ and } f_j^{(s)}(\lambda) > \rho_j^{(s), \text{const}}\}$  is non-empty;
- (B) *Aligned gradients:*  $\nabla f_s^{(s)} \cdot \nabla f_j^{(s)} \geq 0$  near  $\delta_s^{\text{const}}$  (weaker; only guarantees  $f_j^{(s)}(\psi_s^*) \geq f_j^{(s)}(\delta_s^{\text{const}})$ , which is not the same as  $\rho_j^{(s), \text{const}}$ );
- (C) *Pareto-improving region:*  $\mathcal{Q}_j \neq \emptyset$  and the composite optimum lands inside it.

A complete proof set, counter-examples (monotonic OOD, vertex-max, distillation-gap), and the deep asymmetry with Level-1 are in `mof_ood_guarantee_analysis.tex`. The asymmetry's root: Level-1 has a *constructive* fall-back ( $\delta_s^{\text{const}}$ ), but OOD has no analogous safe vertex— $\delta_{k^*}^{\text{const}}$  would be the safe vertex for OOD, but it is poor on the in-domain term, so the composite optimization actively avoids it.

### 4.4 NFT $\beta$ -cancellation

**Theorem 4** ( $\beta$  cancellation). The DiffusionNFT loss (7) satisfies

$$\frac{\partial \mathcal{L}_{\text{NFT}}}{\partial v_{\text{new}}} = A_{\text{max}} \left[ r \frac{\partial \mathcal{M}(v^+)}{\partial v^+} - (1-r) \frac{\partial \mathcal{M}(v^-)}{\partial v^-} \right], \quad (10)$$

and this expression *does not depend on*  $\beta$ . Consequently  $\partial \mathcal{L} / \partial \psi$  is  $\beta$ -independent (the  $\beta$  in the prefactor of  $\mathcal{L}_{\text{NFT}}$  exactly cancels the  $\pm \beta$  chain-rule factors from  $\partial v^\pm / \partial v_{\text{new}}$ ). Only the self-normalization weight  $w^\pm = \text{sg}(\|x_t - \sigma v^\pm - x_0\|_1 / d)$  depends on  $\beta$ , and this enters only as a second-order magnitude rescaling, not direction.

The full derivation is in `nft_beta_analysis.tex`. Practical consequence:  $\beta$  is not a meaningful hyperparameter for MoF; the default  $\beta = 1$  should suffice.



## 4.5 Distillation gap

Stage 2’s MSE (6) is minimized at  $v^\theta = v^{\text{MoF}^*}$  on the support of  $q_\theta$ . Two sources of slack remain:

- **Capacity gap.**  $v^{\text{MoF}^*}$  may not be in the LoRA-parameterized function class.
- **On-policy distribution shift.**  $q_\theta \neq q_{\text{MoF}^*}$ ; even pointwise-perfect MSE on  $q_\theta$  does not guarantee equal trajectory-level rewards (ODE integration amplifies small velocity errors).

**Theorem 5 (Erosion).** Let  $\Delta_{\text{stage1}} = f_j^{(s)}(\psi_s^*) - \rho_j^{(s),\text{const}}$  be the Stage-1 OOD margin and  $\epsilon_{\text{distill}} = f_j^{(s)}(\psi_s^*) - f_j^{(s)}(\theta^*)$  the distillation gap. If  $\epsilon_{\text{distill}} > \Delta_{\text{stage1}}$  then  $f_j^{(s)}(\theta^*) < \rho_j^{(s),\text{const}}$ , *i.e.* the student’s OOD performance is below the best single teacher despite Stage 1 having achieved strict OOD improvement.

**Practical implication:** small Stage-1 margins (the typical regime in RL fine-tuning) are vulnerable to erosion. The Stage-2 student capacity (LoRA rank, training budget) must be large enough that  $\epsilon_{\text{distill}} \ll \Delta_{\text{stage1}}$ .

## 5 Experimental Program

### 5.1 Teacher training (preparation, completed)

We trained three Flow-DPPO teachers from a single base flow model, without classifier-free guidance, on three reward objectives:

| Teacher           | Reward                            | Selected ckpt   | Step | Eval score |
|-------------------|-----------------------------------|-----------------|------|------------|
| teacher_geneval   | GenEval                           | checkpoint-600  | 1200 | 0.958      |
| teacher_ocr       | OCR                               | checkpoint-220  | 440  | 0.953      |
| teacher_pickscore | PickScore (rescaled $\times 26$ ) | checkpoint-1500 | 3000 | 24.128     |

These checkpoints serve as the frozen  $v^{(k)}$  throughout the MoF experiments. Each teacher’s training curve plateaus near these scores; further fine-tuning yields diminishing returns. The selected checkpoints sit *slightly before* the highest-reward step on each curve—a conservative choice that avoids reward-hacking artifacts that often appear late in DPPO fine-tuning (especially for the OCR objective, where the reward continues climbing past the point of legible-text saturation). Figure 1 shows the train and eval reward trajectories for all three teachers.

### 5.2 MoF configurations

We ablate two design axes orthogonally:

- **Mixing module:** `lut`, `lut_simple` (LUT-SA), `time_router`, `adaln_router`, `mlp_router`.
- **RL algorithm:** DiffusionNFT (default), Flow-GRPO.

The reward configuration follows the standard *per-set composite*

$$A(x_b, s_b) = a_{R_{\text{in}}(s_b)}(x_b) + \gamma \cdot \frac{1}{|J_b|} \sum_{j \in J_b} a_j(x_b), \quad J_b = \{j : j \neq R_{\text{in}}(s_b), a_j(x_b) \neq \text{NaN}\}, \quad (11)$$

with three rewards (GenEval, PickScore, OCR), three sources, and `applicable_sources`: `geneval`→[`geneval`] `pick_score`→[`geneval`, `pickscore`, `ocr`] (universal), `ocr`→[`ocr`]. Default  $\gamma = 0.2$ .

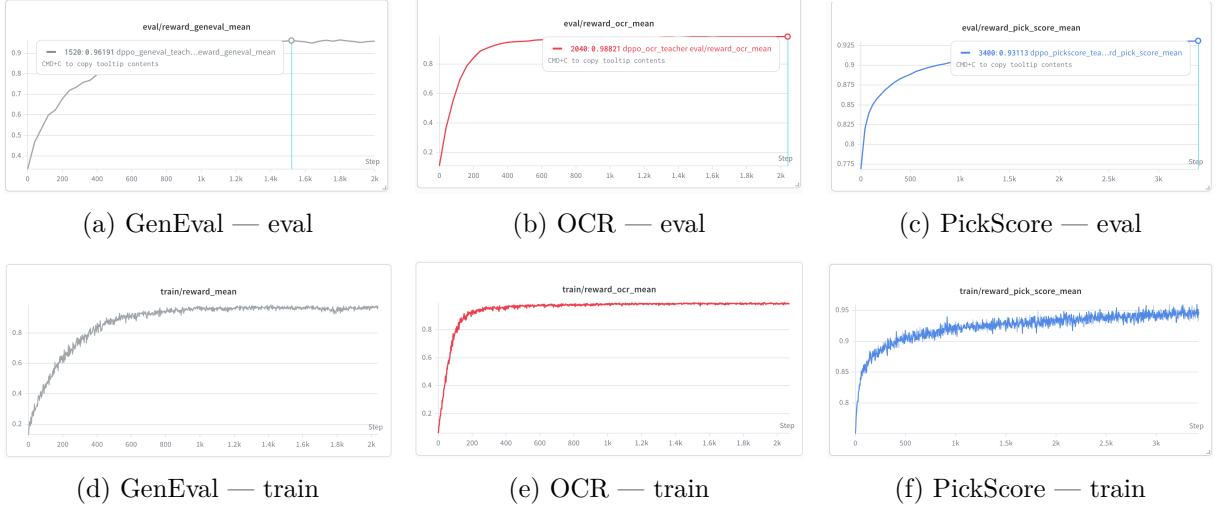


Figure 1: Flow-DPPO teacher training curves (no CFG). **Top row** (a–c): held-out eval reward; **bottom row** (d–f): on-policy training-batch reward. The selected checkpoints used as the frozen  $v^{(k)}$  in MoF are GenEval @ step 1200 ( $R=0.958$ , ckpt-600), OCR @ step 440 ( $R=0.953$ , ckpt-220), and PickScore @ step 3000 ( $R \approx 0.931$  on the rescaled axis, equivalent to 24.128 on the  $\times 26$  unrescaled PickScore). These selections are *conservatively earlier* than the peak-reward steps shown in the tooltips; later checkpoints continue to climb in reward but exhibit qualitative degradation (especially OCR overfitting to text-detection heuristics at the cost of overall image quality). All three curves plateau—further single-objective fine-tuning yields diminishing returns, which is precisely the point at which a multi-teacher mixing approach becomes attractive.

### 5.3 Evaluation

We measure both in-domain and OOD performance:

- In-domain:  $f_s^{(s)}(\theta^*)$  for each  $s$ , vs. the matched single teacher’s eval score above.
- OOD:  $f_j^{(s)}(\theta^*)$  for  $j \neq s$ , vs. each  $f_j^{(s)}(v^{(k)})$  baseline (full  $K \times S$  cross-table).

## 6 Empirical findings and ablation plan

### 6.1 Pathwise vs. PG (verification of Prop. 1)

We run the Stage-2 distillation in three modes: `pathwise_only`, `pg_only`, `both`. Hypothesis: `pathwise_only` matches `both` in final reward; `pg_only` fails to converge or has high-variance training. This validates that Stage 2 is a pure regression problem.

### 6.2 Naive multi-teacher OPD baselines (Step 2)

Two ablations against route-by-source and average-velocity baselines:

- **OPD-route**: target  $v^{\text{tgt}}(x, t, p) = v^{(s(p))}(x, t, p)$ . Hypothesis: in-domain matches teacher (Level-1 trivial), no OOD gain.
- **OPD-average**: target  $v^{\text{tgt}} = \frac{1}{K} \sum_k v^{(k)}$ . Hypothesis: in-domain strictly degrades by Jensen, OOD slightly improves.

These provide the lower-bound reference points for MoF.

### 6.3 MoF main: NFT vs. Flow-GRPO (Step 3)

Run MoF with both RL algorithms, on the LUT and adaLN-router architectures. Hypothesis: NFT converges faster and reaches similar final reward; Flow-GRPO requires more samples but is on-policy. Pick the SOTA per setting.

### 6.4 Temporal analysis of $\lambda^*$

After Stage 1 converges, plot  $\lambda_{k,i,s}$  as a function of timestep  $i$  for each source  $s$ . Hypothesis (informed by analogy with known diffusion phenomena):

- **Early timesteps (high noise):** GenEval-teacher dominates (global layout / object placement).
- **Middle timesteps:** OCR-teacher contributes (text stroke formation).
- **Late timesteps (low noise):** PickScore-teacher dominates (aesthetic refinement).

### 6.5 PickScore-dominance failure mode in NN routers

**Observation:** in preliminary runs with neural routers (Time/adaLN/MLP), the learned router consistently routes *all* samples toward `teacher_pickscore`, even on OCR prompts where PickScore-teacher has poor in-domain performance. LUT and LUT-SA do *not* exhibit this failure.

**Mechanism** (full derivation in `mof_pickscore_dominance_analysis.tex`). Because PickScore is the *only* reward with universal `applicable_sources`, the PickScore signal enters *every* sample’s advantage—once as in-domain (for pickscore samples) and twice via the OOD bonus (for geneval / ocr samples). Total signal coverage on a shared parameter  $\theta$ :

$$F_{\text{pick}} = \frac{1}{3} \cdot 1.0 + \frac{1}{3} \cdot \gamma + \frac{1}{3} \cdot \gamma = 0.467, \quad (12)$$

$$F_{\text{gen}} = \frac{1}{3} \cdot 1.0 = 0.333, \quad (13)$$

$$F_{\text{ocr}} = \frac{1}{3} \cdot 1.0 = 0.333, \quad (14)$$

so the optimum on a shared-parameter router approximately favors PickScore by  $F_{\text{pick}}/F_{\text{ocr}} = 1.4$ . The LUT decouples by source, so each column sees only its own 1.0:0.2 ratio and does not develop dominance.

**Mitigations** (in order of preference):

1. Set  $\gamma = 0$  in router runs (lose OOD bonus entirely);
2. Make PickScore domain-specific (`applicable_sources: [pickscore]`);
3. Add a source-conditioning input to the router so it can decouple internally;
4. Re-weight per-source advantage by inverse signal coverage.

Mitigation 1 is the recommended first step.

### 6.6 $\gamma$ ablation

Grid over  $\gamma \in \{0, 0.05, 0.1, 0.2, 0.5\}$ , on adaLN-router and LUT. Plot in-domain and OOD scores. Hypothesis:

- LUT: small  $\gamma$  improvement is monotone within a Pareto-friendly range.
- Router:  $\gamma = 0$  recovers OCR;  $\gamma > 0$  triggers PickScore-dominance.

This separates the architecture-induced effect from the reward-design effect.

## 6.7 Initialization ablation

Compare `uniform`, `random`, `teacher_biased` (in-domain teacher initialized to bias  $C=2$ ):

- `teacher_biased`: starts near a Level-1-safe  $\delta_s^{\text{const}}$  vertex; should converge fastest with smallest in-domain dip during early training.
- `uniform`: starts at  $1/K$ ; learning trajectory must overcome the OPD-Average mid-point.
- `random`: control for variance.

## 7 Discussion

### 7.1 The asymmetry of in-domain vs. OOD

The framework is fundamentally asymmetric: in-domain Level-1 is a *structural* property of the search space (the safe vertex  $\delta_s^{\text{const}}$  is always there, and is composite-friendly), while OOD strict-improvement requires *additional empirical structure* (temporal complementarity, gradient alignment, or non-empty Pareto region). Practitioners should not promise OOD improvement *a priori*; they should measure it.

### 7.2 When to use which architecture

Briefly:

- **LUT**: best for small  $T, S$  (e.g.  $K = 3, T = 10, S = 3$ ); ablation-friendly because each source/timestep slice is independent.
- **LUT-SA**: diagnostic baseline; identifies whether the  $S$ -axis genuinely helps.
- **Time Router**: continuous- $t$  flexibility (decouple training and inference  $T$ ) without prompt conditioning; recommended for inference-step ablations.
- **adaLN/MLP Routers**: prompt-conditioned mixing; provides the most expressive class but inherits the PickScore-dominance pitfall under default  $\gamma$ .

The decision tree: if your reward configuration has a *universal* reward (covering all sources), prefer LUT or set  $\gamma = 0$  on the router. If all rewards are domain-specific and you want  $T$ -flexibility, the Time Router is a sweet spot.

### 7.3 Limitations

- Gradient magnitude in NFT can become very small ( $\sim 10^{-10}$ ) under LoRA-shared-base teachers, due to softmax mean-subtraction ( $v^{(j)} - \bar{v}$  is a small LoRA residual). Solutions A–D (direct-reward, REINFORCE, residual-space, unnormalized) are catalogued in `mof_per_set_analysis.tex` §13–14.
- Stage 2’s distillation gap is the main practical risk for OOD gains; a multi-round self-distillation refresh of  $v^{\text{MoF}^*}$  may be needed for small margins.
- All theory is conditional on perfect optimization (global maxima of  $F^{(s)}$ ); in practice, RL with a small number of samples may converge to local optima that violate even the Level-1 fallback.

## 8 Conclusion

### Summary

1. OPD distillation in flow-matching reduces to plain velocity MSE because the policy-gradient term is mean-zero (Prop. 1). All distillation in this work is therefore a regression problem.
2. Naive multi-teacher OPD has two failure modes (route-by-source: no OOD gain; averaging: in-domain loss). MoF replaces the rigid target with a learnable convex combination—valid by Prop. 2, optimizable by RL on a tiny parameter set.
3. Theoretically, MoF gives a *constructive* Level-1 in-domain guarantee (Thm. 1), a conditional Level-2 strict improvement (Thm. 2), and *no* general OOD strict-improvement guarantee (Thm. 3); the asymmetry stems from the absence of a composite-friendly OOD safe vertex.
4. The two-stage pipeline (RL→distill) decouples target optimization from deployment cost. NFT is preferred over Flow-GRPO for faster convergence on the small mixing search;  $\beta$  is irrelevant by Thm. 4.
5. In practice, neural routers can fall into a PickScore-dominance attractor when one reward is universally applicable; LUT and LUT-SA escape this by source-decoupling. Set  $\gamma = 0$  or restrict `applicable_sources` to mitigate.

### 8.1 Open problems

1. Designing a Stage-1 objective with a *constructive* OOD fall-back (e.g. a chance-constrained formulation  $\max f_s + \gamma \bar{f}_{\text{OOD}}$  s.t.  $f_j \geq \rho_j^{(s), \text{const}}$ ).
2. Trajectory-level distillation losses that are tighter than pointwise MSE on  $q_\theta$ , reducing Stage-2 erosion.
3. Identifying necessary and sufficient conditions on the teacher ensemble (LoRA rank, training datasets) under which temporal complementarity *provably* holds.
4. Source-conditioned routers with explicit decoupling priors that combine LUT’s robustness with router’s  $T$ -flexibility.

## A Pointers to supporting documents

This notes document consolidates the central narrative; technical derivations live in dedicated companion documents:

| Topic                                                                          | File                                              |
|--------------------------------------------------------------------------------|---------------------------------------------------|
| NFT vs. GRPO loss / gradient comparison                                        | <code>mof_loss_gradient_analysis.tex</code>       |
| $\beta$ cancellation in NFT                                                    | <code>nft_beta_analysis.tex</code>                |
| Per-set $\lambda$ architecture; gradient vanishing analysis; solutions A/B/C/D | <code>mof_per_set_analysis.tex</code>             |
| Reward / advantage pipeline design                                             | <code>mof_reward_design.tex</code>                |
| Continuous router formulation                                                  | <code>mof_continuous_weights.tex</code>           |
| Architecture data-flow (LUT, LUT-SA, Time Router, adaLN, MLP)                  | <code>mof_router_architectures.tex</code>         |
| PickScore-dominance failure mode analysis                                      | <code>mof_pickscore_dominance_analysis.tex</code> |
| OOD strict-improvement (negative theorem; sufficient conditions)               | <code>mof_ood_guarantee_analysis.tex</code>       |
| End-to-end Stage 1 + Stage 2 pipeline; code map                                | <code>mof_two_stage_pipeline.tex</code>           |

## References (informal)

When promoting these notes to a paper, fill in full citations for the following works that are inlined as text above:

- **Flow Matching** (Lipman et al., 2023).
- **Flow-GRPO**, **DiffusionDPO**, **DiffusionNFT** (recent RL-style diffusion fine-tuning works).
- **OPD** (the on-policy distillation line of work, including the multi-teacher route-by-source variant).
- **Flow-DPPO** (the algorithm used to train the three teachers).